

TANMAYA DWIVEDY

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SUMMARY

Motivated Computer Science student with hands-on experience in data analysis, machine learning, and predictive modeling. Skilled in building and deploying data-driven applications using Python, FastAPI, and Streamlit. Possesses strong analytical and problem-solving abilities with an interest in Machine Learning, Deep Learning, and Generative AI technologies.

PROFESSIONAL SKILLS

- Problem Solving
- Analytical Thinking
- Communication
- Team Collaboration

TECHNICAL SKILLS

Machine Learning & Deep Learning: Python, NumPy, Pandas, Scikit-Learn, Machine Learning, Deep Learning, Neural Networks, Transformer Architecture

Generative AI: LLM Fundamentals, Prompt Engineering, Retrieval-Augmented Generation (RAG), LangChain.

Data Analysis: SQL, Exploratory Data Analysis (EDA), Statistical Analysis, Matplotlib, Seaborn

Tools & Platforms: Git, GitHub, AWS EC2, Streamlit, FastAPI

Web Development: HTML, CSS, JavaScript, React (Basics)

PROJECTS

MedBuddy.ML – Heart Disease Risk Prediction System:-

- Developed and deployed an end-to-end machine learning application for predicting heart disease risk by performing data cleaning, exploratory data analysis, and outlier detection on healthcare datasets.
- Built and evaluated supervised machine learning models using robust validation techniques and domain-relevant metrics such as Recall and F1-score.
- Converted experimental notebooks into modular **Python scripts** and deployed the solution using a **FastAPI** backend and Streamlit frontend on **AWS EC2**, enabling real-time predictions through a user-friendly interface.

House Price Prediction System:-

- Developed an end-to-end regression-based machine learning application to predict housing prices using structured real estate data.
- Performed exploratory data analysis using Matplotlib and Seaborn to identify feature relationships, trends, and outliers.
- Built a robust preprocessing and modeling pipeline using ColumnTransformer with One-Hot Encoding, Standard Scaling, and **Simple Imputer**.
- Evaluated multiple regression models using MAE, RMSE, and R^2 metrics and selected the best-performing model based on validation results.
- Applied **Cross-Validation** and **Hyperparameter Tuning** to improve model performance and generalization capability.

Customer Churn Prediction System:-

- Built a supervised machine learning model to predict customer churn by analyzing customer behavioral and service usage data.
- Performed data preprocessing and feature engineering, handled class imbalance using SMOTE, and evaluated models using 5-fold cross-validation.
- Implemented and compared multiple classification algorithms and selected the best-performing model based on validation accuracy.
- Finalized and optimized the **Random Forest** classifier to support data-driven customer retention strategies.

EDUCATION

Bachelor of Technology in Computer Science:-

Bachelor of Technology in Computer Science Kalinga Institute of Industrial Technology (KIIT)

CGPA: 8.8/10(88%) (Till Semester 6)

Completed academic and personal projects involving data analysis, machine learning model development, and statistical techniques using Python. Actively participated in technical clubs and coding competitions, strengthening analytical thinking and problem-solving skills.

Intermediate:-

BridgeWell Global School, BBSR, Odisha.

Matriculation:-

DAV Public School NTPC/TTPS, Thermal/Talcher, Odisha.

July 2023 - April 2027

June 2021 - April 2023

ADDITIONAL INFORMATION

Languages: English, Hindi, Odia.

Certifications:

- Machine Learning Bootcamp (2026)
- The Complete SQL Bootcamp (2026)
- Data Structures and Algorithms (DSA) using Java